Review plan · for the teacher

# How agents work: the loop, tools and memory: unit check

AI agents · Level 3 · Unit check

| Time | Materials |
| --- | --- |
| 45 min | Board, chalk, paper. No devices needed. |

## Learning goals

- Inside the loop
- How it picks a tool
- Memory: the context window
- Teaching it a new tool

## Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

## Discussion

- Which idea will you use first with your class?
- Which of Darsi Tiny's mistakes surprised you most?
- What would you check before trusting an agent at school?

## Exit ticket

- One thing I now understand about how agents work, and how I'll explain it to my pupils

**Try it at school tomorrow**
Pick the activity from this unit your class needs most (the paper strip, the new wordings, the ten-word window or the new signal), run it this week, and tell a colleague how it went.

Quiz sheet · for each team

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Team _____ Date _____

**1** **In an agent's text, who writes each tool's result?**

◯ The person who asked ◯ The model, from what it learned ◯ The app, from the tool

**2** **Put the app's loop in order.**

☐ Check the call

☐ Send the text again, until the model answers

☐ Run the tool and add its result

☐ Send the text so far to the model

☐ The model writes a tool call

**3** **A call that copies a class name wrongly can safely run, because the tool will report an error anyway.**

◯ True ◯ False

**4** **Match each slip to the check that catches it.**

| | | |
|---|---|---|
| 9Z written as 9C | • • • • | The tool must be on |
| A reminder at 25:00 | • • • • | Every name must come from the request |
| A call to a tool that is switched off | • • • • | No step before it may have failed |
| A reminder set after the timetable tool returned an error | • • • • | Every detail must have the right shape |

**5** **What does this program print?**

```
steps = 0
while steps < 5:
    steps = steps + 1
print(steps)
```

◯ 4 ◯ Nothing: it never stops ◯ 5

**6** **Which are true about an agent's context window? Choose all that apply.**

☐ Darsi Tiny's holds 384 tokens ☐ The model can look up turns that fell out

☐ It is the model's only memory in a conversation ☐ What falls out of it is gone for the model

☐ Big cloud models never forget anything

**7** **Complete the sentences about the teach lab.**

With no old conversations mixed in, Darsi Tiny learned the bus tool but kept only _____ of 8 old skills. With 48 mixed in, it kept _____ of 8.

**Word bank:** 5 · 1 · 0 · 8

**8** **A step limit stops a loop in which the model keeps calling tools and never answers.**

◯ True ◯ False

**9** **The school wants its agent to understand room codes like "Lab 2". What should come first?**

◯ Fine-tune it tonight on twenty examples ◯ Switch to a bigger cloud model

◯ Try a clearer instruction or a room-codes tool, with a check in code

**10** **This loop should stop as soon as the model answers. Which line is the mistake?**

Circle the line with the mistake. Write it correctly:

```
for step in range(5):
    part = model(text)
    if is_answer(part):
        continue
    text = text + part + run_tool(part)
```

Answers · for the teacher

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**1** **The app, from the tool**

The model writes plans, calls and answers, then stops after a call.

**2** **Top to bottom: 3, 5, 4, 1, 2**

The app sends the text, the model writes a call, the app checks it before anything runs, then runs the tool and adds the result, and goes round again until an answer.

**3** **False**

Not every slip gives an error: a wrong class that exists returns the wrong lesson, and a reminder or message just acts.

**4** **9Z written as 9C — Every name must come from the request**
**A reminder at 25:00 — Every detail must have the right shape**
**A call to a tool that is switched off — The tool must be on**
**A reminder set after the timetable tool returned an error — No step before it may have failed**

Each is one of the simple rules of a checker in code: names from the person's words, the right shape, tools that are on, and no building on a failed step.

**5** **5**

steps goes up by one each time round: 1, 2, 3, 4, 5.

**6** **It is the model's only memory in a conversation; What falls out of it is gone for the model; Darsi Tiny's holds 384 tokens**

The window is the only memory, and what falls out is gone.

**7** **With no old conversations mixed in, Darsi Tiny learned the bus tool but kept only 1 of 8 old skills. With 48 mixed in, it kept 8 of 8.**

New examples alone made it forget: old skills fell from 8 of 8 to 1 of 8.

**8** **True**

After a fixed number of calls the app stops the loop and tells the person, instead of running forever and using up battery, data and time.

**9** **Try a clearer instruction or a room-codes tool, with a check in code**

An instruction or a tool changes nothing else in the model.

**10** **Line 4:** `break`

continue skips to the next turn instead of leaving the loop, so the model is asked again after it has already answered.